

PARTICULAR SPECIFICATION P8

Architectural Works

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NOTE: THIS DOCUMENT SUPERCEDES THE ORIGINAL VERSION OF THE P8 – ARCHITECTURAL WORKS AT THE CLOSE OF TENDER. THE CHANGES HAS BEEN HIGHLIGHTED IN YELLOW.

P8.1 INTRODUCTION

P8.1.1 Overview

- a) This section describes the minimum requirements for architectural and building works. This Specification shall be read in conjunction with all other sections of the Contract.
- b) This Specification cover the following facilities. The list of rooms and architectural elements are described in the Room Data Sheet and are summarised below:
 - (i) Sidestream Nitrogen Removal Facility
 - 1) Electrical Switchrooms and Transformer Bays;
 - 2) Cable chambers;
 - 3) Process XLV and Server Room, Non-process XLV;
 - 4) Battery Room;
 - 5) UPS Room;
 - 6) Operations Room;
 - 7) Gas Suppression Cylinder Room;
 - 8) Pipe Gallery;
 - 9) Stairs;
 - 10) Drainage System; and
 - 11) Doors and shutters.
 - (ii) Digestion Pre-treatment Facility
 - 1) Maintenance platforms and stairs at Thermal Hydrolysis Process (THP) Area.
- c) The provision of store room, meeting room, pantry, locker room, changing room, toilet, office, laboratory, multipurpose room are not required. The provision of any other rooms that are not described in the Room Data Sheet shall be subjected to S.O.'s approval.

P8.2 ARCHITECTURAL WORKS

P8.2.1 General

- a) This section of the Specification relates to all architectural building Works required under the Contract. The particular requirements shall complement the general requirements in the General Specifications. Where there is a contradiction between the details in the Particular Specification and the General Specification, or between different sections of the Particular Specification, or Drawings, the Contractor shall seek clarification in writing from the S.O. before proceeding with provision of the affected items.

P8.2.2 Sustainability

- a) All architectural materials used (except where specifically exempted by the S.O.) shall be environmentally friendly and certified by approved local certification authorities. They shall be of requisite sustainability quality to achieve a rating of 'Excellent' under category NRB 3-2 ('Sustainable Products') of the BCA Green Mark Scheme.
- b) Green Cements shall be used in place of Ordinary Portland Cement, and Recycled Concrete Aggregates or Washed Copper Slag from approved sources shall replace coarse and fine aggregates.

P8.2.3 Fire escape from C4A Biosolids Building Exit Staircases

- a) In the design of the THP Area, the Contractor shall ensure 1.2m wide escape path to be maintained for occupants to escape from C4A Biosolids Building exit staircases to safe area.
- b) The escape path shall be free of obstruction and unhindered by any flammable material, gases or steam from the THP equipment.

P8.2.4 Fire Rated Walls and Doors

- a) The fire resistance ratings of all walls and doors are generally indicated in the Room Data Sheets and the Door Schedules. Where none are indicated, the following shall apply:
 - (i) All walls above ground shall be of minimum 2-hour fire rating.
 - (ii) All fire doors shall have the same fire rating as the walls in which they are set.
- b) All ironmongery on fire doors shall be of the same fire rating as the doors on which they are to be installed, including the entire door system which includes and is not limited to the door frame, closers, etc. and shall have the necessary fire test certificates and Certificates of Conformity (COC) which corresponds to the entire door system and which the warranty must not have lapsed.
- c) All fire rated doors shall be fitted with self-closing device and fire rated doors shall comply with the latest Singapore Fire Code.

- d) Certificate of Conformity (COC), detailed fire tests certified reports and the necessary fire safety reports of shall be furnished to S.O. for all fire rated products and materials upon completion.

P8.2.5 Fire Access Panels or Windows

- a) All Fire Access Panels shall comply with the requirements of the latest Singapore Fire Code.
- b) For the avoidance of doubt, the fire access panels shall be fully openable from the exterior of the building and installed with the necessary red triangle in weather-proof exterior grade vinyl adhesive film, as well as signage text on the interior stating "FIRE ACCESS PANEL – DO NOT OBSTRUCT". The text 'FIRE-FIGHTING ACCESS – DO NOT OBSTRUCT' in etched lettering 50mm tall shall be indicated on stainless steel signage on the inside of the access panels.
- c) When any FAP Panel or window is required to be locked due to safety reason or any other operational requirements, a break glass box to store key shall be provided both on the internal and external side.
- d) All fire access panels at locations with cladding shall be weather sealed and shall be contiguous and compatible with the cladding system. Such access panels in the cladding shall afford direct and unencumbered access into the building, without secondary access panels.

P8.2.6 Fire Fighting Equipment

- a) All relevant fire safety provisions and fire-fighting equipment required shall comply with the requirements of the latest Singapore Fire Code, Fire Safety Guidelines and the corresponding Codes of Practice and Standards.
- b) The Contractor shall provide wall mounted ABC type fire extinguishers to all areas based requisite Singapore Fire Code requirements. The fire extinguishers shall be of the correct type(s) and installed at the height correct height with clear signage 'Fire Extinguisher' with contrasting background, it shall comply with the latest Singapore Fire Code and the applicable standards (SS 578 : 2012 formerly CP55) shall inspected and have valid labels. Where the extinguishers are located within cabinetry, clear signage must be included. Electrical LV and HV Switch Room shall be provided with wall mounted carbon dioxide (CO₂) type fire extinguishers.
- c) All fire extinguishers shall be installed with framed stainless steel signage panels measuring 210mm by 300mm cantilevered from the wall using concealed fixings, with no exposed screws, fasteners or bolts. The extinguisher symbol and text shall be clearly etched on the sign.

- d) All hose reels shall have standard hose reel signage with the text 'TURN ON THE INLET VALVE BEFORE RUNNING OUT THE HOSE' in text 50mm high. Where hose reels are housed in cabinets, the hose reel cabinet door shall have a sign 210mm by 300mm with the hose reel symbol and text 'HOSEREEL CABINET'.
- e) Fire hydrants are to be located along the fire engine accessway according to Singapore Fire Code requirements. At any point along the fire engine accessway, the fire engine should not be greater than 50m away from the fire hydrant. Siting and type of fire hydrants to comply with SS CP 29: Code of Practice for Fire Hydrant Systems and Hose reels.

P8.2.7 Ironmongery

- a) The Contractor shall provide ironmongery in accordance with the General Specifications. All locks, latch sets and accessories shall be stainless steel SS 316.
- b) To ensure consistency across the entire project, all permanent door locks shall be from the same supplier as determined for the entire Tuas Water Reclamation Plant and shall be from the same system to allow site-wide compatibility, including flexibility for a site-wide master keying system, if required. The lock cylinders will be provided by the C6B contractor for installation by the Contractor.
- c) The Contractor shall provide construction key systems for all doors to the following rooms. The construction key system shall be installed in accordance to the General Specifications G8 Architectural:
 - (i) All electrical rooms;
 - (ii) XLV rooms;
 - (iii) Control rooms;
 - (iv) UPS and battery rooms; and
 - (v) Transformer Bays.
- d) Unless indicated, all doors shall be installed with closers. All fire door ironmongery shall be tested in accordance with the requirements of the Singapore Fire Code and shall have the necessary fire ratings and Certificates of Conformity (COC). The hardware for the closers will be provided by the C6B contractor for installation by the Contractor.
- e) The Contractor shall allow for a system of temporary construction phase locksets to ensure security during construction prior to provision of the permanent locksets.
- f) There shall be no door knobs in this entire Contract.

P8.2.8 Railings

- a) Where a railing or panel is the only installation between a building occupant and a fall of more than 1m, the railing or panel shall be designed to withstand a horizontal

loading in accordance with BS 6399 : Part 1 and/or SS EN 1991 (and/or Section 8 of BS 6180 where applicable).

- b) The Contractor shall submit a Professional Engineer's endorsement indicating compliance with the above (with structural calculations where required) for the purposes of obtaining BCA TOP and CSC.
- c) For the avoidance of doubt, all openings in the floors, whether left uncovered, or covered with stainless steel SS 316 grilles, stainless steel SS 316 gratings, or pre-cast slabs, and which are above the slab below by a vertical distance of more than 1m, shall be provided with railings to the same design and specification as above, whether or not they are drawn in the models. Where such railings enclose a pipe, duct or valve requiring maintenance access, the railing shall be designed such that it has provision of a lockable gate. Where such railings enclose an opening to be used for hoisting, the railing shall be designed such that it may be removed without undue difficulty for access during hoisting but shall not be easily removable outside of hoisting periods. Locations of fixed and removable railings shall be subjected to S.O. approval during construction stage.

P8.2.9 Staircases

- a) All staircase treads shall be fitted with non-slip grooved colour contrasting nosing tiles between 50mm and 65mm in width. The nosing tiles shall be yellow in colour; samples shall be submitted to the S.O. for approval.
- b) All staircase floor finishes shall comply with the current edition of Singapore Standard (SS) 485 regarding slip resistance.
- c) All staircases shall have a minimum internal headroom of 2200mm (measured vertically from the pitch line). All access to staircase doors, lift lobby doors and smoke stop lobby doors shall be unencumbered, notwithstanding the placement of pumps and other equipment on the slab. It shall be the responsibility of the Contractor to coordinate the access paths to the staircases so that no item of equipment shall obstruct this free and unencumbered access.
- d) Any flammable gas pipes that are located adjacent to/ abutting to/ within close proximity to the internal or external exit passageways, internal or external exit staircases and fire escape routes shall be encased with 4-hour proprietary fire rated board system.

P8.2.10 Steel Platforms, Steel Staircases and Pipe Racks

- a) All internal and external use steel platforms (including railings) and steel staircases (including maintenance access ladders) designated for maintenance access and exit access shall be fireproofed with intumescent paint with a minimum 2-hour fire rating. Certificate of Conformity (COC) shall be furnished and submitted to S.O. upon completion of works.
- b) Contractor shall provide internal and external access platforms and railings to comply with BCA requirements and the latest Singapore Fire Code and prevailing standards for the required loading capacity (max. load 540kg), clear width of walkway, fire escape distances and other prevailing authorities' compliances etc. Contractor shall submit to S.O. the designs for approval, prior to PE endorsement and PE submissions to authorities.
- c) Pipe racks shall be provided to be able to take on the weight of pipes and contents including the loading weight when accessed for maintenance. All pipe racks shall be fireproofed with intumescent paint with a minimum of 2-hour fire rating as specified in Drawings. Certificate of Conformity (COC) shall be furnished and submitted to S.O. upon completion of works. Headroom height clearance signages for pipes or services crossing roads shall be provided.
- d) Within the Digestion Pre-treatment Facility, the structural steel works that are integral part of the THP Units shall be provided with coating as per THP manufacturer's standard and to have a minimum design lives of 25 years.

P8.2.11 Doors

- a) All doors shall be in painted finish to S.O.'s approval.
- b) All external doors to be set back from face of external wall by 50mm. 20mm deep 'v' groove to be provided for rainwater to drip away from top edge of external door. All external doors to be provided with stainless steel SS 316 drip edge at top edge. All external doors that open to driveways to be installed with vision panels that comply to COC of the door system.
- c) All doors shall be fitted with 'Push' and 'Pull' stainless steel plates measuring 40mm by 120mm on both sides of the door.
- d) Rain protection awnings (not more than 2m projected from the external building wall) shall be provided at the external doors and external openable windows. Rain protection awnings shall be of sturdy materials and construction, awnings shall have the capacity to withstand the wind load in the designated installation areas. The fixed awning's types, material samples and installation details shall be submitted to S.O. for approval.
- e) All doors which need to be installed with biometric readers shall be coordinated with Building Services, specialist's consultants and Other contractors.

P8.2.12 Signage

- a) The provision and installation of all signages for fire safety and fire-fighting requirements (including those not mentioned in this Specification) shall comply with the latest Singapore Fire Code, Fire Safety Guidelines and the corresponding Codes of Practices/Standards.
- b) Proposed artwork of all signage shall be submitted to the S.O. for approval. The Contractor shall propose the text/graphics/artwork on the signs to be confirmed by the S.O., subject to limitations of size.
- c) All signages (design, types of materials, quantity and mounting locations) shall be submitted to the S.O. for approval.
- d) Complete schedules of all room/riser door name signs/directional, complete with names and numbers, shall be submitted to the S.O. for approval.
- e) All exit doors shall be fitted with luminous strips around the door frame on the side facing occupants who are travelling in the direction of escape.
- f) Except where otherwise indicated, all signage shall be in minimum 3mm thick marine grade stainless steel plates, with etched lettering generally in black but with other colours as well for highlights and emphasis. Where signs are to be cantilevered, the Contractor shall allow for text and graphics on both sides of the signs with adequate fastening to adjacent walls. Where there are no adjacent structures to affix the signage, additional structures/attachments shall be included to align the signage in a discernible height to the viewer. Corners and edges of signage to be treated to prevent injuries where required.
- g) All emergency exit signage shall comply with the Singapore Fire Code and SS 263, SS 508 and SS 563 and General Specifications G5 Electrical.
- h) Each Dry Riser and Breeching Inlet shall have standard signage installed at locations that are clearly visible for fire fighter and visible from fire engine accessway.
- i) All statutory and safety precautionary signages shall be provided and submitted to S.O. for prior approval, as required for compliance with relevant statutory requirements (FSSD, BCA, LTA, NParks, NEA, MOM/WSHA etc.).
- j) All room doors shall be fitted with Room Name signage, with the room name and number etched. Each Room Name signage shall measure 75mm by 200mm.
- k) All riser doors shall be fitted with Riser Name signage, with the riser name and number etched. Each Riser Name sign shall measure 75mm by 200mm.

- l) All staircase landings linking directly to building storeys (but not including intermediate landings) shall be fitted with staircase landing signage measuring a minimum of 300mm by 300mm each, indicating the staircase number, the storey of the landing and whether or not there is roof access. All exit staircase signage shall comply with the latest Singapore Fire Code.
- m) All buildings shall be fitted with wall-mounted directional signs, that may be flat on the walls or cantilevered from them with concealed fixings. Cantilevered signs shall be framed and may contain information on both sides. Cantilevered signs will be attached to adjacent structures, where there no adjacent structures, signage may be self-supported with additional structures where required. These directional signs shall measure a minimum of 840mm by 600mm, and the number of signs to be provided shall be based on the ratio of one such sign for every 100m² of floor space in the building.
- n) Fire escape plans shall be provided and displayed in common lobbies such that they are easily viewable by the building occupants passing through these common areas. The fire escape plan shall have legible letterings and the fire escape routes made clear to the viewer. It shall clearly show the layout of the floor in the correct orientation and highlight the escape routes (in relation to viewer's location), escape corridors and exit staircases using appropriate colours, directional signs and words. All fire escape plans shall comply with the latest Singapore Fire Code.
- o) Signages for the Fire Engine Accessway shall be provided along the peripheral and comply with the signage requirement as stipulated in the Singapore Fire Code.
- p) Signages for Floor Loading capacity shall be provided at each storey. The signages shall be located within the buildings/structures.
- q) Headroom height clearance signages for pipes or services crossing roads shall be provided.
- r) Hazard markings and signages to be provided for low headroom services or objects that have less than 2m headroom clearance from the floor finished level.
- s) Safety signages to be provided on deep tank covers at Sidestream Nitrogen Removal Building.

P8.2.13 External Aluminium Cladding, Roofing and Panels (Aluminium Finishes)

- a) External aluminium cladding shall consist of perforated aluminium panels with standard perforation or proposed 50% opening, whichever conforms to performance requirements. The Contractor shall provide details of the proposed aluminium cladding and support substructure framing systems and interface details to the S.O. for approval. The cladding shall be from the standard product range of a cladding manufacturer with requisite track records. The Contractor shall provide the aluminium cladding and support substructure framing systems and shall be responsible for completing the detailed design and interfacing with the main building, including PE endorsement and necessary authority submissions and approvals. All coatings for aluminium claddings/roofing/panels shall comply with the following standards:
 - (i) Powder coating shall meet AAMA 2604 Standard;
 - (ii) PVDF coating shall meet AAMA 2605 Standard; and
 - (iii) Anodizing shall comply with AAMA 611 Standard.
- b) Modular and/or non-modular perforated cladding panels and louvres shall be provided with matching joints and trimmings and finished in manufacturer's PVDF coatings. Louvres to be finished with an inhibitive thermo-cured primer, with a dry film thickness average of 5 to 7 microns. Followed by a thermo-cured fluorocarbon coating with a minimum dry film thickness of 25 microns. All aluminium panels shall be thoroughly cleaned, etched and given a chromated conversion pre-treatment before application of the Fluorocarbon (PVDF) coating. The coating shall receive a bake cycle in accordance with the paint manufacturer's specification. All finishing procedures shall be one continuous operation in the approved plant of the manufacturer's applicator. Manufacturer to furnish a warranty for a period of 10 years for the Fluorocarbon (PVDF) coating.
- c) The Contractor shall provide certified test results by a recognised testing laboratory or agency in accordance with specified test methods for each assembly.
- d) Panel Deflection shall be limited to no greater than 1/720 of panel span applied to positive loading only, as tested in accordance with ASTM E330.
- e) Fire Access Panels (FAP) shall be co-ordinated with the aluminium cladding and windows.
- f) Ventilation louvres shall be co-ordinated with the aluminium claddings and windows, they shall be contiguous and compatible with the cladding system without affecting the performance of the weatherproof louvers.

- g) The aluminium cladding panels/screens/louvres shall be provided in at least three different shades of colours (to be submitted of S.O. approval), to be selected and approved by S.O.. Contractor shall allow for on-site mock up to ascertain the aesthetic and interface details for approval.
- h) All aluminium claddings and roofing, at the factory, will be applied with a self-cleaning coating, (Titanium Dioxide or equivalent) which will be bonded to the cladding or external surface at a molecular level. The surface of the aluminium cladding to be cleaned prior to the application of the self-cleaning coating. Curing time to be negligible. The application of the self-cleaning coating will be applied by the specialist installer, at the factory, to the aluminium cladding prior to delivery on site, and protected with plastic covering. There will be a minimum of 2 coats applied to the aluminium cladding. The Contractor will submit PSB Test JIS Z 2801 : 2000 or equivalent for Antimicrobial products – Test for antimicrobial activity and efficacy to determine effectiveness of the self-cleaning coating and to provide a warranty of 10 years.

P8.2.14 Drainage Systems

- a) Internal floor drainage system at each facility shall be provided.
- b) All floor finishes (including concrete floors) sloped to fall to the nearest scupper drains/floor traps. All scupper drains to be provided with stainless steel SS 316 grating. Contractor shall submit sample to the S.O. for approval.
- c) All drain covers shall be stainless steel SS 316 in various sizes and configuration according to the required drainage layout. Contractor shall refer to the LTA standards for drainage and manholes. Drain's and manhole's cover, sizing and support and be submitted to S.O. for approval.
- d) The roof drainage system for the project shall be based on a siphonic drainage system.
- e) The Contractor shall allow for all costs for the successful implementation of a siphonic rainwater drainage system, including connection to the stormwater drains at grade. The Contractor shall allow for a siphonic rainwater drainage system that will cater to a 100-year return period storm event with a peak intensity increase of 25%.
- f) The Contractor shall construct the roof/roof insulation/screed such that slopes of the roof slabs are graded toward the rainwater outlet to correspond with the Contractor's proposed siphonic rainwater drainage system.
- g) For Siphonic Rainwater Outlets, all key components shall be manufactured from non-corrosive metallic materials such as Marine Grade Stainless Steel SS 316, Aluminium Alloy or equivalent.

- h) For Pipe Conveyance Systems, the Contractor shall design each siphonic system to:
 - i) Ensure that all pipe materials and pipe system support structure as specified are capable of withstanding the forces and reactions so applied to them due to:
 - 1) Changes in the dynamic flow of the system;
 - 2) Thermal and seismic effects;
 - 3) Static Load;
 - 4) Hydrostatic pressure; and
 - 5) Deflections in the structural elements of the building directly supporting the pipework, if identified.
 - i) The pipe material for the project shall be of 6 bar Pressure UPVC Pipe System. Where a potential blockage can occur in a rainwater drainage system, the pipe system shall be capable of resisting the hydrostatic forces from such blockage. Where this pressure exceeds the pressure rating of the pipe system designed for dynamic pressure under siphonic full-bore flow, the specification of the pipe system under hydrostatic should be made from a suitable metallic pipe material.

P8.2.15 Raised Cavity Flooring

- a) All raised flooring shall comply with the requirements of the latest Singapore Fire Code. Complete Certificates of Conformity (COC) for entire systems shall be submitted for all raised flooring systems proposed to be installed.
- b) Contractor shall note that the raised flooring in the Sidestream Nitrogen Removal Facility XLV/Server Rooms may be installed in phases to be coordinated with the C5A contractor's installation of cabling and equipment. Contractor shall allow for all co-ordination with all suppliers/contractors laying materials and services within the raised floor cavity. Contractor shall allow for full installation of the raised floor system during construction of the Sidestream Nitrogen Removal Facility, full dismantling and full re-installation at a date, to be co-ordinated with the S.O., and shall allow for all mobilisation, supervision, inspection and rectification costs.
- c) Raised floor to be in heavy duty vinyl finish to S.O.'s approval.

P8.2.16 Brickwork

- a) All brick walls greater than 1.0m high shall be tied to a concrete column or wall. The anchor bolts shall be designed by the Contractor to avoid steel reinforcement in the column or wall and to not affect the structures strength or stability. Details of the proposed fixing arrangements shall be submitted to the S.O. for approval.
- b) Only brick work not receiving a permanent covering of plaster or rendering, located above ground level and visible shall be fair facing brickwork.

P8.2.17 Concrete Block

- (a) Block shall be hard, even in shape, square with true arises, well mature, of a dense and of standard sizes 390 x 190 x 100mm thick or 190 x 90 x 100mm thick.
- (b) All concrete blocks shall be from an approved manufacturer and shall be in accordance with BS EN 15435.
- (c) Load-bearing blocks shall have a crushing stress of not less than 5.6 MN/m² and non-load-bearing blocks shall have a crushing stress of not less than 3.6 MN/m².
- (d) Quarter, half, three quarter size and special corner blocks shall be used for bonding where applicable.
- (e) Design Criteria
 - (i) Dry Density: 490-500kg/m³
 - (ii) Min. Compressive Strength: 2.5MPa
 - (iii) Thermal Performance: R-value
 - 1) 100mm thickness: 0.77m²K/w
 - 2) 200mm thickness: 1.54m²K/w
- (f) Fire Rating Wall Assemblies
 - (i) 100mm and 200mm thickness: 2 hours SPRING tested and labelled.
- (g) All block walls to be plastered and painted finish.

P8.2.18 Internal and External Building Finishes

- a) All stainless steel material shall be SS 316 and in hairline finish. Thicknesses of stainless steel material shall be sturdy and with adequate strength to withstand the designated/required loading weight.
- b) Internal floor, wall and ceiling finishes are detailed in the Building Finishes section of the Room Data Sheet provided as an Appendix to this Specification.
- c) Floor screed shall be installed with a minimum thickness of 20 mm and laid to fall.
- d) Benching shall be provided with a minimum thickness of 20 mm to avoid feather edges.
- e) The internal floor finished level shall be higher than the external finished levels. Internal floor finish shall be sloped to fall whenever applicable, external floor finish levels and external concrete aprons shall be sloped to the external drain or road whichever is applicable. The internal floor finished level for 1st storey shall be +5.300 SHD.

- f) Bare wall and bare concrete floor finishes to all rooms/areas except for rooms that will be used and occupied for operations shall be in plaster and paint finished. Bare wall and bare concrete floor finishes shall be level, even and gradient to fall. All skim coat to be 3mm thick, unless otherwise indicated.
- g) Walls to be plastered and painted shall be with recessed PVC groove lines in the plaster, unless otherwise indicated. All exterior walls shall be painted with external coatings suitable for the environment and all interior walls shall have skim coats. All exterior paintings and coatings shall comply with BCA's requirements for daylight reflectance and tests certificates shall be submitted to S.O. for approval prior to procurement of paintings and coatings. Specifications and colour samples for all paintings and coatings shall be submitted to obtain prior approval from S.O..
- h) In certain instances (e.g., at flat roof locations) there is a substantial difference between the structural floor level and the finished floor level. Generally, this is to cater to rooftop waterproofing and insulation thickness, as well as grading requirements for the efficient flow of rainwater across the surface. The Contractor shall price to allow the designated finished floor levels to be met.
- i) Damp-proof membrane, vapour barriers and waterproofing layers shall be provided at the foundations, 1st Storey floor slabs and basement/semi-basement slabs, vertical walls/building walls abutting to open grade/soil areas exposed to weather etc. Application of damp-proof membrane, vapour barriers and waterproofing layers shall comply with BCA standards and other applicable guidelines/standards.
- j) Expansion and Structural joints at the roofs and floors (concrete floor slabs, floor finishes) to have adequate upturn kerb and to have manufacturer's flashing and waterproofing to prevent seepage of water into the building. Samples to be to S.O.'s approval. Expansions and structural joints at other (non-concrete) roofs shall be applicable and meet the selected materials requirements. Expansions and structural joints for concrete slabs or any other applicable areas shall meet structural requirements and be submitted for structural approval.

P8.2.19 Sidestream Nitrogen Removal Facility Operations Room

- a) The Operations Room shall be located within the Sidestream Nitrogen Removal Facility on level 1. The Architectural Works for the room are indicated in the Room Data Sheet and include but are not limited to the provision of:
 - (i) Reinforced concrete slabs, columns, beams, lintels and roof slab;
 - (ii) Brickwork panels tied to concrete columns;
 - (iii) Doors, windows and louvres;

- (iv) Floor screeds;
 - (v) Building finishes; and
 - (vi) Motorised operated roller blinds for all windows inclusive of black out blinds. Steel roller tubes and fabric colour fastness of 7 Blue Scale to AS 2001.4.21.
- b) The consoles and loose furniture shall be supplied by C5A, allowance shall be included for co-ordination and assistance to facilitate installation. Quantity and model for all other built-in fitments (cabinets, joinery works) are to be submitted for S.O. approval.

P8.2.20 Electrical Switchrooms and Transformer Bays

- a) The Architectural Works for the Electrical Switchrooms are indicated in the Room Data Sheet and include but are not limited to the provision of:
- (i) Provision of reinforced concrete base slab, columns, walls, and roof slabs;
 - (ii) Provision of mass concrete benching, screeds and blinding;
 - (iii) Provision of brickwork panels tied to concrete columns.
 - (iv) Provision of access gates;
 - (v) Provision of transformer plinths;
 - (vi) Provision of granite chips inside structure;
 - (vii) Provision of drainage openings from roof of structure to site drainage system; and
 - (viii) Provision of double roof at areas below AHU units. Double roofs to be provided with weep holes for drainage.
- b) Roof penetration to the switchrooms and electrical rooms are not allowed. All penetration shall be via dog-houses and shall be submitted to S.O. for approval.
- c) Access to each transformer bay shall be via a lockable galvanised mild steel grill double leaf gate of minimum width for access, operations and maintenance. The Contractor shall supply a padlock, suitable for outdoor use and six (6) sets of keys for securing the transformer gates.

P8.2.21 Roof and Peripheral Access

- a) Access to the roof is for maintenance only. Unless fitted with suitable temporary edge safety rails during construction, all personnel accessing the roof shall wear a suitable full body harness with lanyard and lifeline (fall-arrest system) and shall anchor themselves to the structure. Contractor shall provide suitable stainless steel SS 316 safety line and anchor points bolted to the roof slab at 5m centres along the access and along the complete perimeter of the roof.
- b) The Contractor shall provide an access ladder with safety cage to the roof.
- c) All access maintenance catwalk shall be fitted with 1m high railings. Fire hose reels of 30m direct distance coverage shall be provided along the catwalk.
- d) All maintenance access platforms, ladders, galleries, lifelines (fall-arrest system) etc. shall comply with WSHA requirements and guidelines. All cat ladders, maintenance accessways that lead to the roof and locations of high-risk high elevation areas for maintenance and operations purposes to be installed with fall-arrest system conforming to authority requirements to S.O.'s approval.
- e) All safety barriers and railings heights and designs shall comply with the prevailing regulations and applicable guidelines (BCA, FSSD, WSHA etc.).

P8.3 APPENDIX

- a) APPENDIX P8A – Room Data Sheet
- b) APPENDIX P8B – Door Schedule

END OF SECTION